

Counterfactual Analysis for Structural Dynamic Discrete Choice Models

A Partially Identified Entry and Exit Model

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Idea of the paper

- Kalouptsi, Kitamura, Lima, and Souza-Rodrigues (2026), “Counterfactual Analysis for Structural Dynamic Discrete Choice Models,” *Review of Economic Studies*.
- Discrete choice data only identifies differences in utilities.
- The primitive parameters and counterfactuals can't be point-identified, in general.
- How much can we learn about counterfactual outcomes from set-identified model?

Goals

- Validate replication code for the paper's dynamic discrete-choice model and partial-identification procedure.
- Visualize the trade-off between entry-cost and marginal-cost subsidies under partial identification.

What I did

- Let AI read the paper.
- Ask AI to explain the methodology using a minimal example to check their understanding.
 - The paper is clear, so AI understood well. But some explanation was hard to follow, so I checked AI's understanding from several aspects.
- Discuss the detail of toy-model and let AI write the code.
 - This required several rounds of back-and-forth communication.
 - Ended up just using the paper's example.
- Discuss visualization of counterfactuals.
 - AI is bad at visualization.

Model

A four-state model isolates entry and exit

- Replicate a dynamic discrete-choice model with payoff parameters that are not point identified.
- Carry the estimated payoff set into policy counterfactuals.
- A firm chooses $a_t \in \{0, 1\}$ in state $x_t = (k_t, w_t)$.
 - $k_t = a_{t-1}$ records the previous action.
 - $w_t \in \{L, H\}$ is the observed profit state.
- The profit state follows the known transition matrix P .
- The next lagged-action state is $k_{t+1} = a_t$.

Flow payoffs create entry and exit margins

Previous action	Inactive, $a_t = 0$	Active, $a_t = 1$
$k_t = 0$	0	$\pi_w - r_w - e_w$
$k_t = 1$	s_w	$\pi_w - r_w$

- e_w : entry cost
- r_w : running cost
- s_w : scrap value received upon exit

Value iteration delivers the CCPs

The choice-specific values are

$$v_a(k, w) = u(k, w, a) + \beta E[V(a, w') \mid w].$$

Type-I extreme-value shocks imply

$$V(k, w) = \log [\exp\{v_0(k, w)\} + \exp\{v_1(k, w)\}],$$

$$p_{kw} = \frac{\exp(v_1(k, w))}{\exp(v_0(k, w)) + \exp(v_1(k, w))}.$$

Identification

Observed choices do not reveal one payoff vector

- The data identify four CCPs, $p_{kw} = \Pr(a = 1 \mid k, w)$, and the transition matrix P .
- Logit inversion converts each CCP into a value difference:

$$d_{kw} = \log \left[\frac{p_{kw}}{1 - p_{kw}} \right] = v_1(k, w) - v_0(k, w).$$

- The value differences restrict payoffs, but they do not separately determine every s_w , e_w , and r_w .

The first restriction removes continuation values

For a fixed profit state, both equations contain the same continuation term C_w :

$$d_{0w} = \pi_w - r_w - e_w + C_w,$$

$$d_{1w} = \pi_w - r_w - s_w + C_w.$$

Subtracting the equations gives

$$d_{1w} - d_{0w} = e_w - s_w.$$

- The CCPs identify entry cost relative to scrap value, not their separate levels.

Inactive paths identify the value-function gap

- If the firm chooses $a = 0$, both lagged-action states move to $k' = 0$ next period.
- Their continuation values therefore cancel when the two inactive paths are compared.
- The logit inclusive-value identity gives

$$V(1, w) - V(0, w) = s_w + \log \left[\frac{1 - p_{0w}}{1 - p_{1w}} \right] = s_w + A_w(p).$$

- This value gap recovers the continuation term that appears in the first pair of equations.

A candidate scrap value determines the other costs

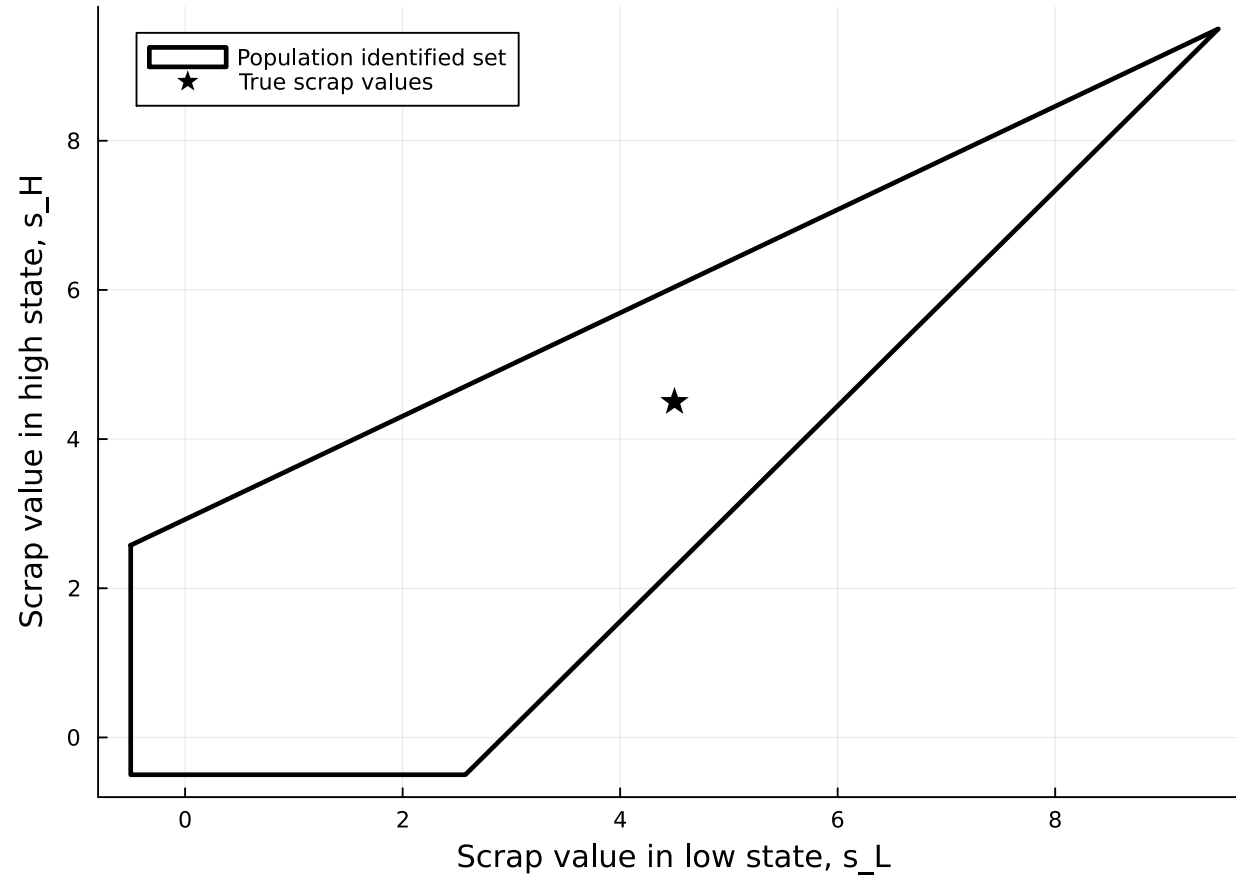
For any candidate $s = (s_L, s_H)$, the two restrictions imply

$$e_w(s; p) = s_w + d_{1w} - d_{0w},$$

$$r_w(s; p) = \pi_w - e_w(s; p) - d_{0w} + \beta \sum_{w'} P_{ww'} [s_{w'} + A_{w'}(p)].$$

- Once s is selected, the data determine $e(s; p)$ and $r(s; p)$.
- The restrictions $e_w(s; p) \geq 0$ and $r_w(s; p) \geq 0$ select a polygon of admissible scrap values.

Identification delivers a polygon, not a point



Estimation

Estimation mirrors the population argument

1. Simulate actions and profit transitions.
2. Estimate the four CCPs and the profit transition matrix.
3. Replace (p, P) with $(\hat{p}, \hat{P})$ in the payoff restrictions.
4. Construct the estimated polygon in (s_L, s_H) .

The sample objective stacks the four restrictions:

$$\hat{Q}(\theta) = \hat{m}(\theta)' \hat{m}(\theta).$$

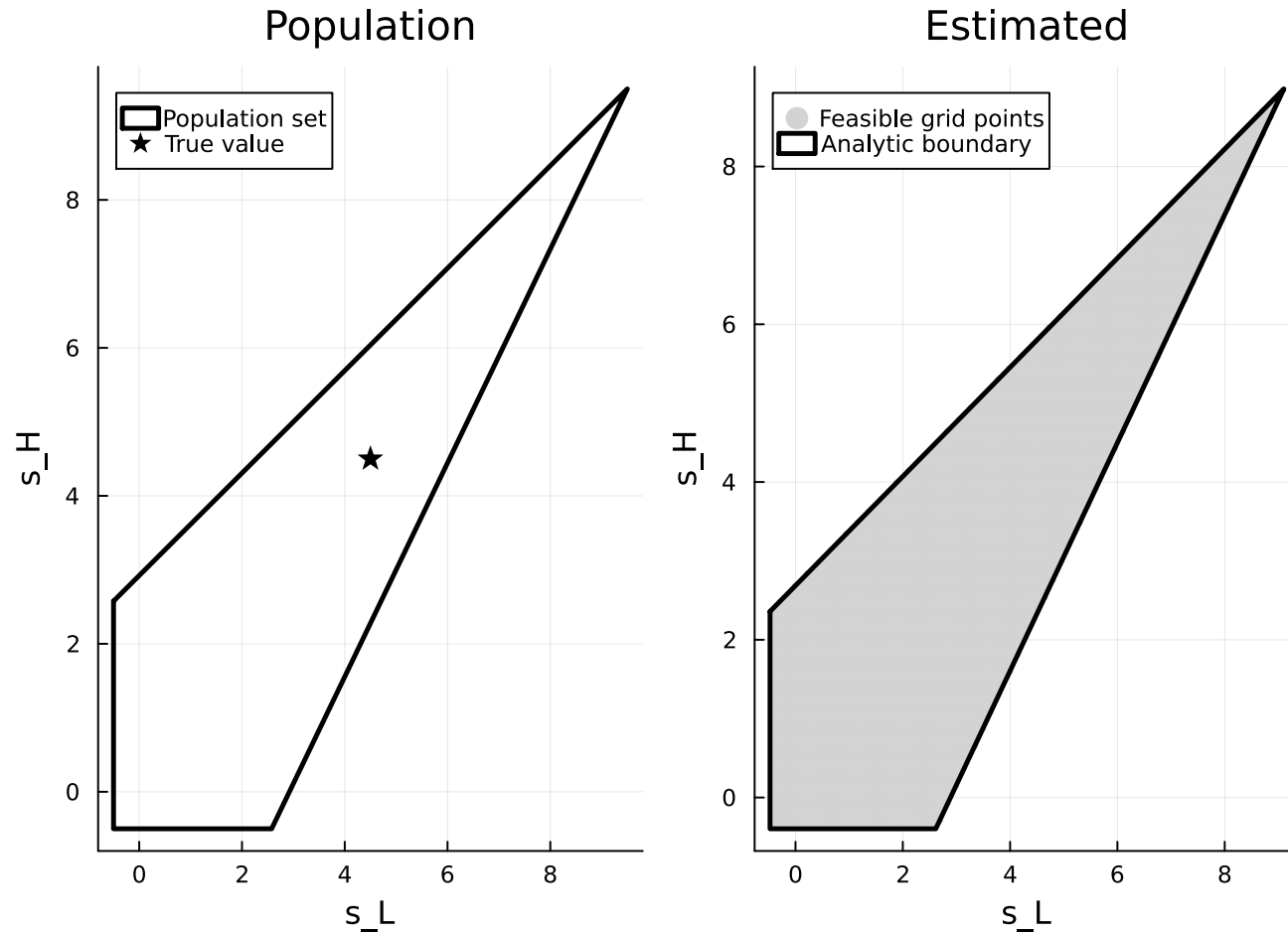
The simulated data recover the four CCPs

k_t	Profit state	Observations	Active choices	Active share
0	Low	4232	3024	0.715
0	High	2504	2393	0.956
1	Low	15684	12562	0.801
1	High	17580	17048	0.97

Estimated and population CCPs:

k_t	Profit state	$\hat{p}_{kw}$	p_{kw}
0	Low	0.715	0.714
0	High	0.956	0.951
1	Low	0.801	0.805
1	High	0.97	0.97

The estimated object is also a polygon



Counterfactual Analysis

Policy changes costs, not the baseline environment

- The policy grid uses $\tau_E, \tau_M \in [0, 0.5]$.
- An entry subsidy lowers the cost paid when an inactive firm enters:

$$e_w^{cf} = (1 - \tau_E)e_w.$$

- A marginal-cost subsidy lowers running cost whenever the firm operates:

$$r_w^{cf} = (1 - \tau_M)r_w.$$

- The exercise holds π , β , P , and the shock distribution fixed.

Each policy requires a new dynamic solution

For each $\theta \in \hat{\Theta}_I$ and each (τ_E, τ_M) :

1. Apply the subsidies to flow payoffs.
2. Solve the Bellman equation for p_{kw}^{cf} .
3. Combine p_{kw}^{cf} with P to construct T^{cf} .
4. Solve $\mu^{cf} = \mu^{cf} T^{cf}$ for the stationary distribution.

The implied long-run active share is

$$\mu_1^{cf} = \sum_{k,w} \mu_{kw}^{cf} p_{kw}^{cf}.$$

Outcomes compare activity, spending, and efficiency

The activity effect is the percentage change from the same model's baseline:

$$R_{\mu} = 100 \frac{\mu_1^{cf} - \mu_1^0}{\mu_1^0}.$$

Long-run entry and operating expenditures are

$$G_E = \sum_w \tau_E e_w \mu_{0w}^{cf} p_{0w}^{cf},$$

$$G_M = \sum_{k,w} \tau_M r_w \mu_{kw}^{cf} p_{kw}^{cf}.$$

Cost-effectiveness is $CE = R_{\mu} / (G_E + G_M)$ when expenditure is positive.

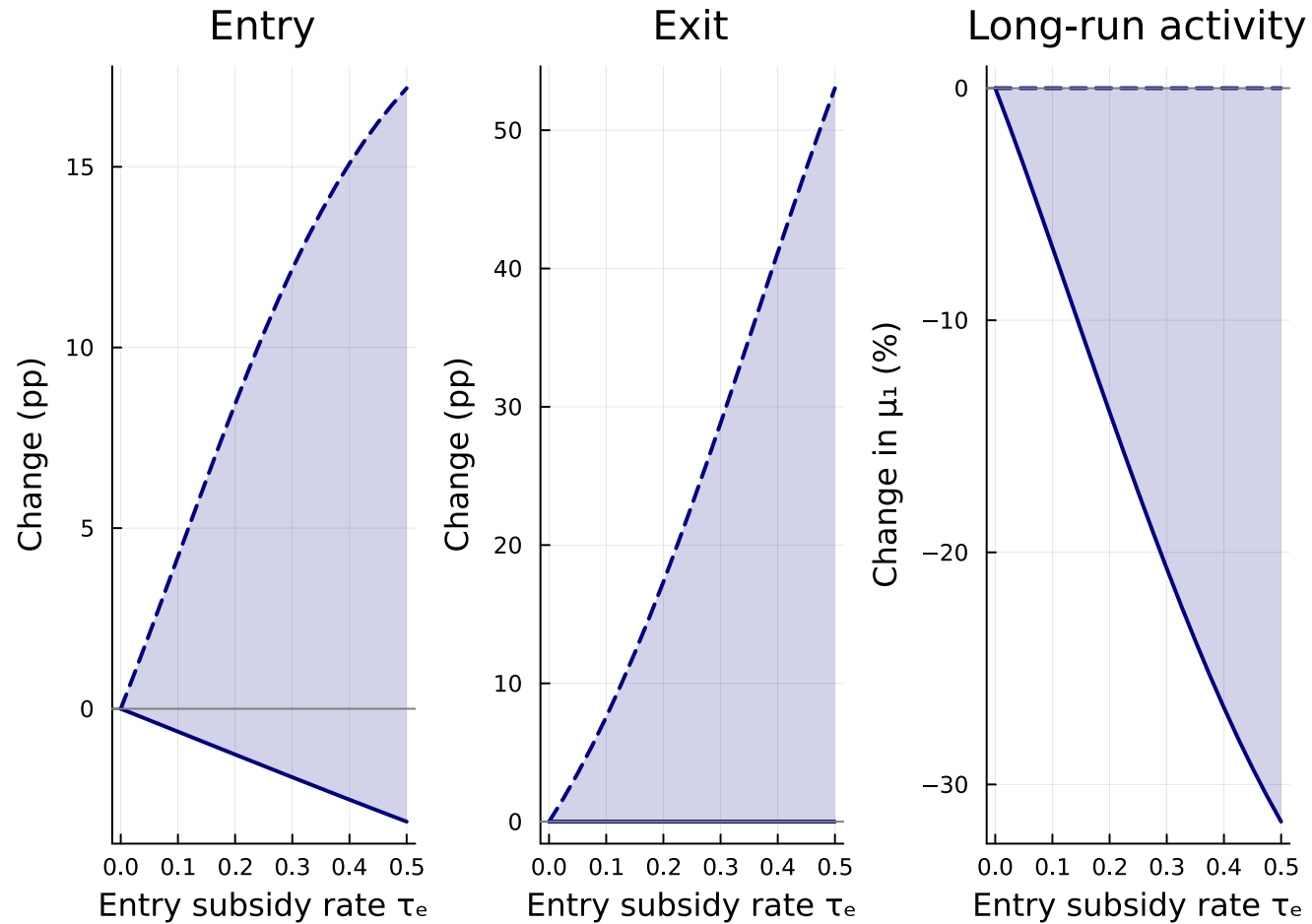
Policy outcomes remain set identified

All payoff vectors in $\hat{\Theta}_I$ match baseline behavior, but they can imply different policy responses. For any outcome Y , define

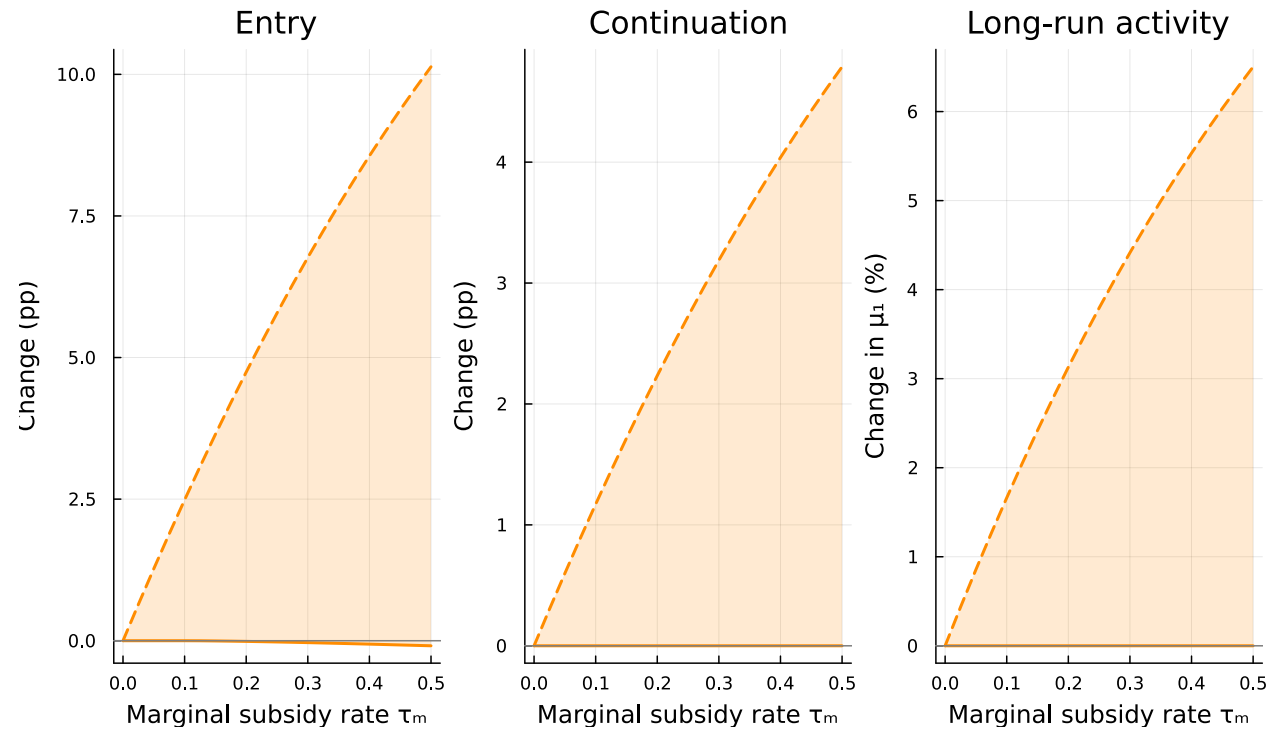
$$\mathcal{C}_Y(\tau_E, \tau_M) = \{Y(\theta, \tau_E, \tau_M) : \theta \in \hat{\Theta}_I\}.$$

- Y_L is the smallest element of $\mathcal{C}_Y$; Y_U is the largest.
- The figures report these bounds, not a single selected parameter vector.

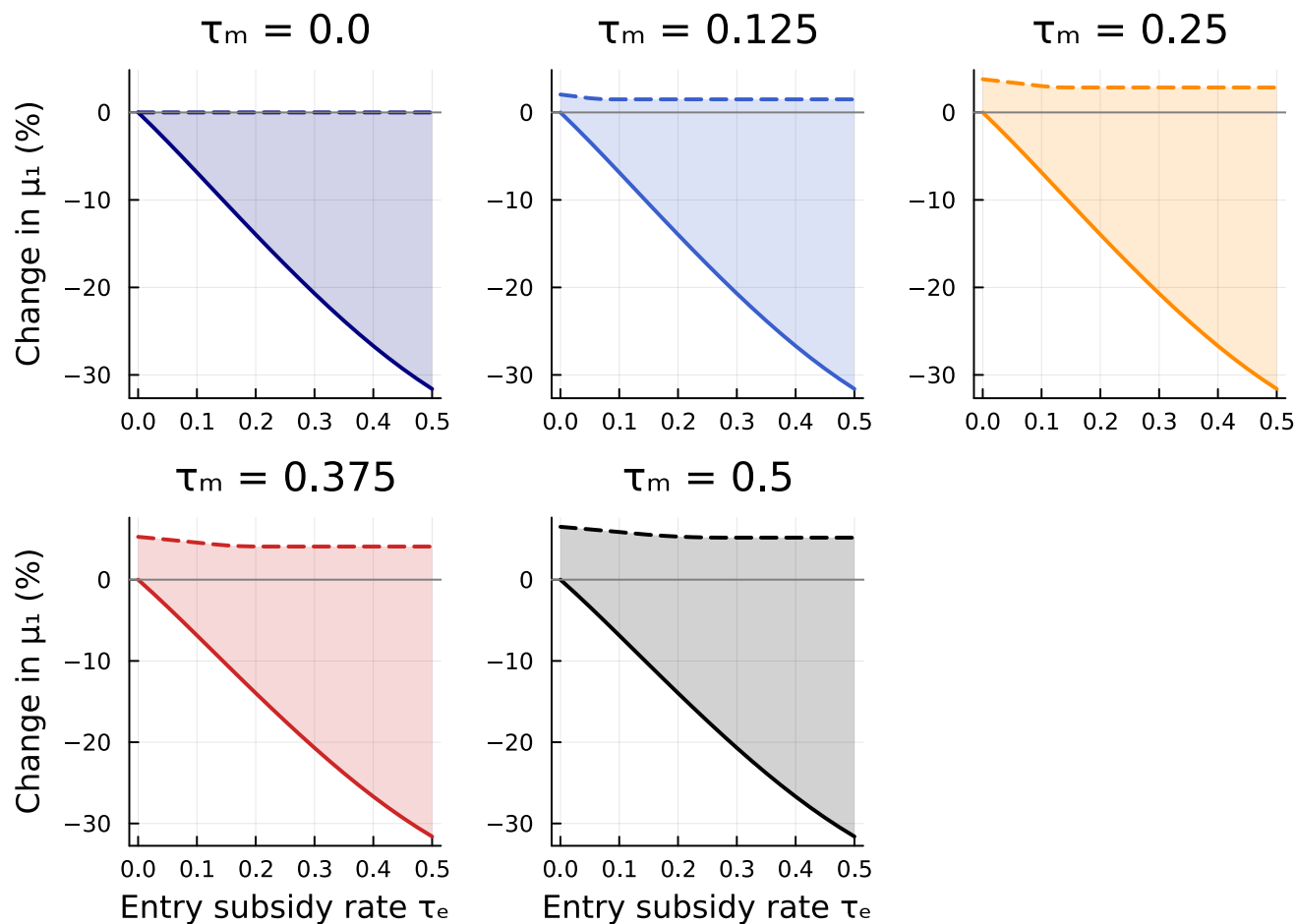
Entry support can raise exit



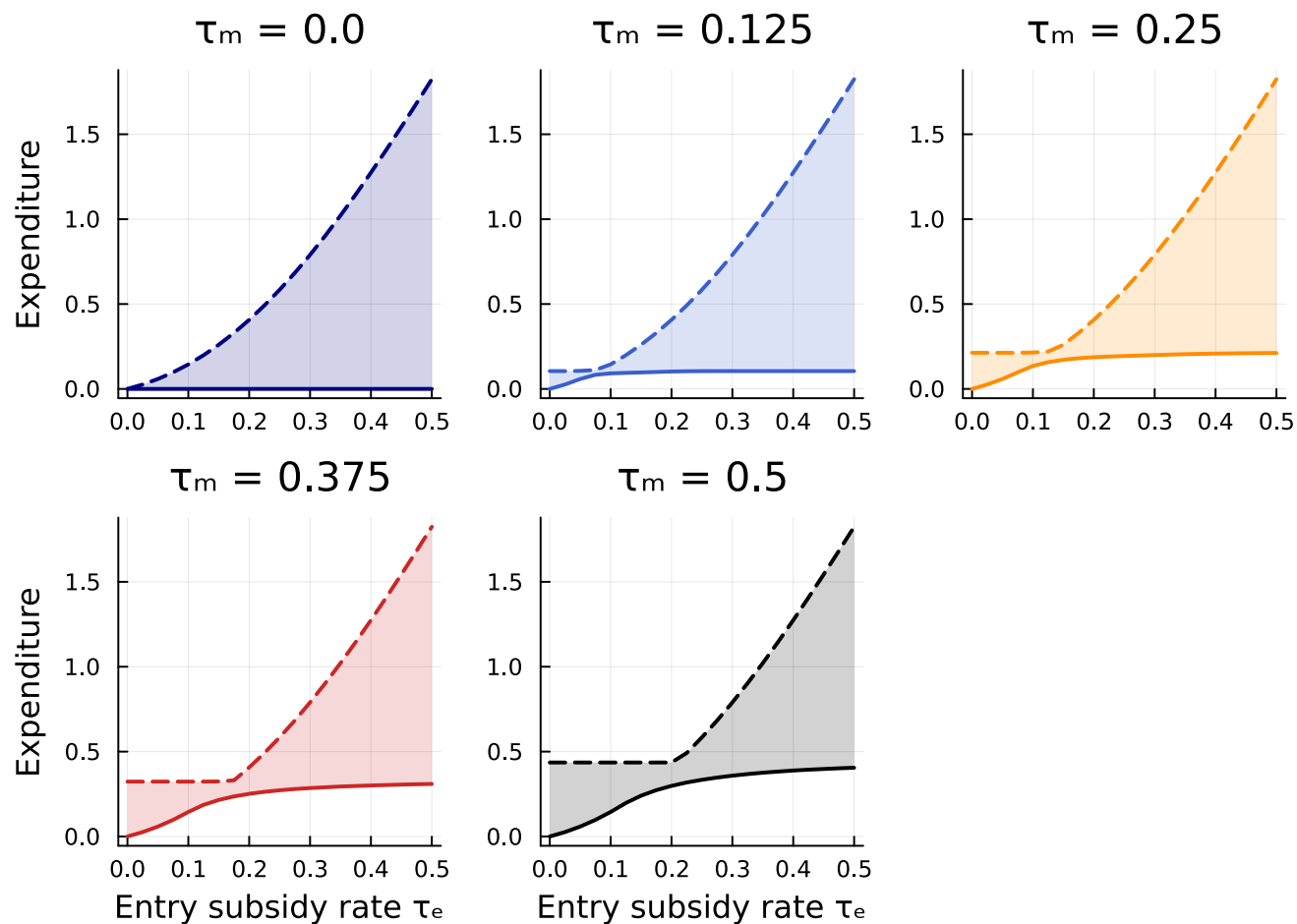
Operating support raises continuation



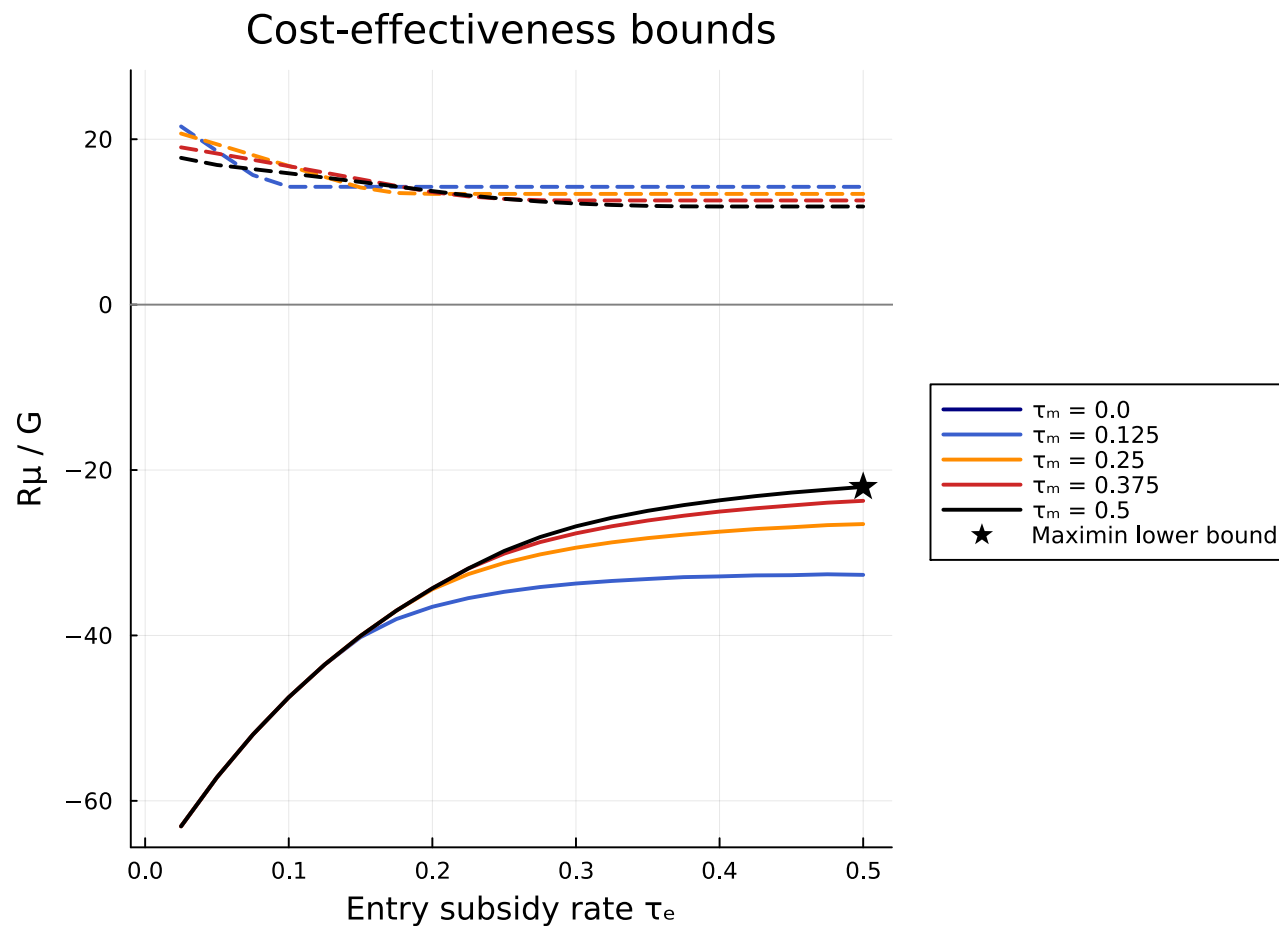
Mixed policies produce set-valued activity effects



Expenditure depends on policy timing



Cost-effectiveness is itself partially identified



The identified set changes policy conclusions

- Baseline CCPs determine a payoff set rather than one structural model.
- Every counterfactual therefore re-solves all admissible models.
- Entry support can raise both entry and exit, so long-run activity may fall.
- Marginal-cost support raises continuation but requires recurring payments.
- Policy comparisons must report both expenditure and outcome bounds.